ASSIGNMENT-10.5

Lab 10 – Code Review and Quality: Using AI to Improve Code

Quality and Readability

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Task 1: Improving Variable & Function Names Original Code

def f(a, b): return

a + b print(f(10,

20))

CODE:

def add_numbers(first_number, second_number):

"""

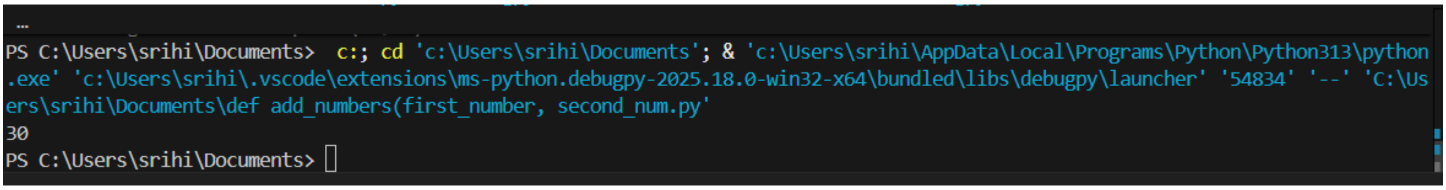
Returns the sum of two numbers.

"""

return first_number + second_number

result = add_numbers(10, 20) print(result)

OUTPUT



```
PS C:\Users\srihi\Documents>  c:; cd 'c:\Users\srihi\Documents'; & 'c:\Users\srihi\AppData\Local\Programs\Python\Python313\python
.exe' 'c:\Users\srihi\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '54834' '--' 'C:\Us
ers\srihi\Documents\def add_numbers(first_number, second_num.py'
30
PS C:\Users\srihi\Documents>
```

Task Description #2 – Missing Error Handling

Task: Use AI to add proper error handling.

Sample Input Code: def

divide(a, b): return a / b

print(divide(10, 0))

Expected Output:

• Code with exception handling and clear error messages

CODE :

def divide_numbers(dividend, divisor):

    """

    Divides two numbers safely.

    """

try:

        return dividend / divisor   except

ZeroDivisionError:

        return "Error: Division by zero is not allowed."

except TypeError: return "Error: Please enter valid

numbers."

result = divide_numbers(10, 0) print(result)

OUTPUT:

```
PS C:\Users\srihi\Documents>  c:; cd 'c:\Users\srihi\Documents'; & 'c:\Users\srihi\AppData\Local\Programs\Python\Python313\python
.exe' 'c:\Users\srihi\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '62114' '--' 'C:\Us
ers\srihi\Documents\Untitled-1.py'
Error: Division by zero is not allowed.
PS C:\Users\srihi\Documents>
```

Task Description #3: Student Marks Processing System

The following program calculates total, average, and grade of a student, but it has poor readability, style issues, and no error handling.

marks=[78,85,90,66,88] t=0

for i in marks:

t=t+i

a=t/len(marks) if

a>=90: print("A")

elif a>=75:

print("B") elif

a>=60: print("C")

else:

print("F")

Task:

- Use AI to refactor the code to follow PEP 8 standards.
- Add meaningful variable names, functions, and comments.
- Add basic input validation and documentation.

CODE :

```
def calculate_average(marks_list):
    """
    Calculates the average of student marks.

    Args:
        marks_list (list): List of student marks
    Returns:
        float: Average marks
    """    if not marks_list:        raise ValueError("Marks list
cannot be empty.")

    return sum(marks_list) / len(marks_list)
```

```
def determine_grade(average_marks):

    """

    Determines grade based on average marks.

    """

    if average_marks >= 90:

        return "A"     elif

average_marks >= 75:

return "B"     elif average_marks

>= 60:

        return "C"     else:

        return "F"



def main():

    """

    Main function to process student marks.

    """

    marks = [78, 85, 90, 66, 88]

    try:

        average = calculate_average(marks)        grade =

determine_grade(average)


        print(f"Average Marks: {average:.2f}")

print(f"Grade: {grade}")


    except ValueError as error:        print(f"Error:

{error}")



if __name__ == "__main__":
```

main()

OUTPUT:

```
PS C:\Users\srihi\Documents>  c:; cd 'c:\Users\srihi\Documents'; & 'c:\Users\srihi\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\srihi\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '62436' '--' 'C:\Users\srihi\Documents\Untitled-1.py'
Average Marks: 81.40
Grade: B
PS C:\Users\srihi\Documents>
```

Task Description #4: Use AI to add docstrings and inline comments to the

following function. def factorial(n):

result = 1 for i in

range(1,n+1):

result *= i

return result **CODE :**

```
def factorial(number):
    """
    Calculates the factorial of a given number.
    """
    if number < 0:        raise ValueError("Negative numbers not
allowed.")

    result = 1

    for i in range(1, number + 1):
        result *= i

    return result


if __name__ == "__main__":
```

number = int(input("Enter a number: "))   print("Factorial:",

factorial(number))

OUTPUT:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\srihi\Documents>  c:; cd 'c:\Users\srihi\Documents'; & 'c:\Users\srihi\AppData\Local\Programs\Python\Python313\python
.exe' 'c:\Users\srihi\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58292' '--' 'C:\Us
ers\srihi\Documents\task 4.py'
Enter a number: 6
Factorial: 720
PS C:\Users\srihi\Documents> \
```

Task Description #5: Password Validation System (Enhanced) The following

Python program validates a password using only a minimum length check,

which is insufficient for real-world security requirements.

pwd = input("Enter password: ") if

len(pwd) >= 8: print("Strong") else:

print("Weak")

CODE : import re

def is_strong_password(password):

"""

Checks whether the given password meets security standards.

Rules:

- Minimum 8 characters
- At least one uppercase letter
- At least one lowercase letter
- At least one digit
- At least one special character

```
    Args:        password (str): User-entered
password

    Returns:        bool: True if strong, False
otherwise
    """

    if len(password) < 8:
return False

    if not re.search(r"[A-Z]", password):
        return False
    if not re.search(r"[a-z]", password):
        return False

    if not re.search(r"[0-9]", password):        return
False

    if not re.search(r"[!@#$%^&*()_+{}[\]:;<>,.?~\\-]", password):        return
False

    return True


def main():
    """
    Main function for password validation.
    """
    user_password = input("Enter password: ")
```

```
    if is_strong_password(user_password):

        print("Password is Strong ")     else:

        print("Password is Weak ")

print("It must contain:")        print("- At

least 8 characters")        print("- One

uppercase letter")        print("- One

lowercase letter")        print("- One digit")

print("- One special character") if

__name__ == "__main__":


    main()
```

OUTPUT

```
PS C:\Users\srihi> & C:/Users/srihi/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/srihi/Documents/import re.py"
Enter password: Srihitha@2005
Password is Strong ✅
PS C:\Users\srihi>
```